

UNIT -3

Introduction to Cloud Computing

Cloud computing has transformed how organizations build, deploy, and scale technology. Instead of owning physical infrastructure, users access computing resources over the internet on demand.

- Eliminates the need for physical servers and upfront hardware costs.
- Enables on-demand access to computing power and storage.
- Supports scalability, flexibility, and cost efficiency.

Problem That Cloud Computing Solves

Before the cloud, if a company wanted to launch a website or an application, it had to follow a slow and expensive process:

The "Old Way" (On-Premises)

- Companies had to purchase physical servers, storage devices, and networking equipment.
- This required a large upfront investment known as Capital Expenditure (CapEx).
- Peak usage had to be estimated in advance, often leading to overprovisioning and wasted resources.
- Procuring and setting up new servers could take weeks or even months.

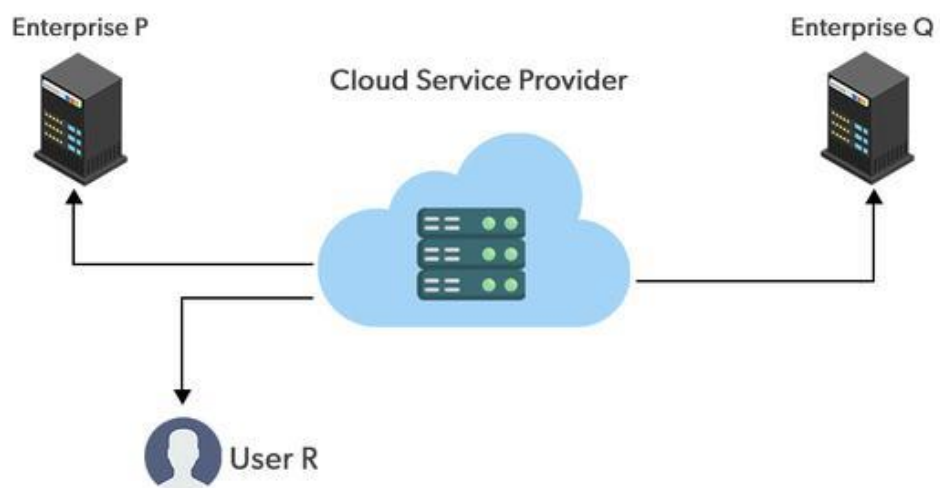
The "New Way" (Cloud Computing)

- Hardware is no longer purchased; computing resources are rented from cloud providers such as AWS, Google, or Microsoft.
- Costs shift from a large upfront investment to a pay as you go Operational Expenditure model.
- Resources can be provisioned instantly without long waiting periods.
- Eliminates overprovisioning, guesswork, and resource wastage.

Cloud Computing Deployment Models

1. Public Cloud

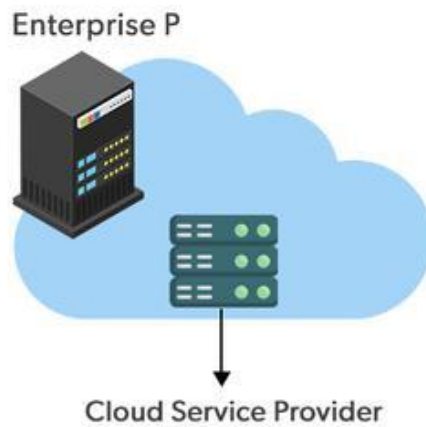
- Public clouds are managed by third parties which provide cloud services over the internet to the public, these services are available as pay-as-you-go billing models.
- They offer solutions for minimizing IT infrastructure costs and become a good option for handling peak loads on the local infrastructure. Public clouds are the go-to option for small enterprises, which can start their businesses without large upfront investments by completely relying on public infrastructure for their IT needs.
- The fundamental characteristics of public clouds are multitenancy. A public cloud is meant to serve multiple users, not a single customer. A user requires a virtual computing environment that is separated, and most likely isolated, from other users.



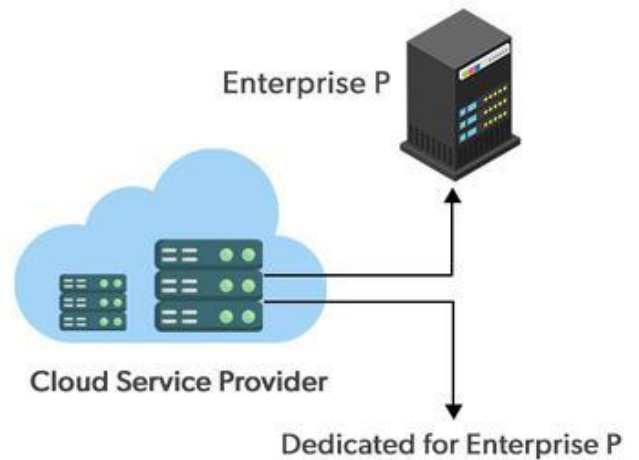
2. Private cloud

- Private clouds are distributed systems that work on private infrastructure and provide the users with dynamic provisioning of computing resources. Instead of a pay-as-you-go model in private clouds, there could be other schemes that manage the usage of the cloud and proportionally billing of the different departments or sections of an enterprise. Private cloud providers are HP Data Centres, Ubuntu, Elastic-Private cloud, Microsoft, etc.

On premise Private cloud



Externally hosted Private cloud



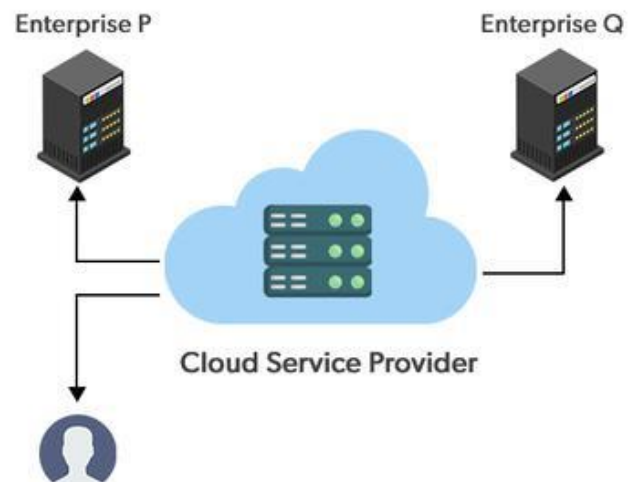
3. Hybrid cloud

- A hybrid cloud is a heterogeneous distributed system formed by combining facilities of the public cloud and private cloud. For this reason, they are also called heterogeneous clouds.
- A major drawback of private deployments is the inability to scale on-demand and efficiently address peak loads. Here public clouds are needed. Hence, a hybrid cloud takes advantage of both public and private clouds.
- Examples: [AWS Outposts](#), Azure Stack, Google Anthos, IBM Cloud Satellite, Oracle Cloud at Customer

Private cloud

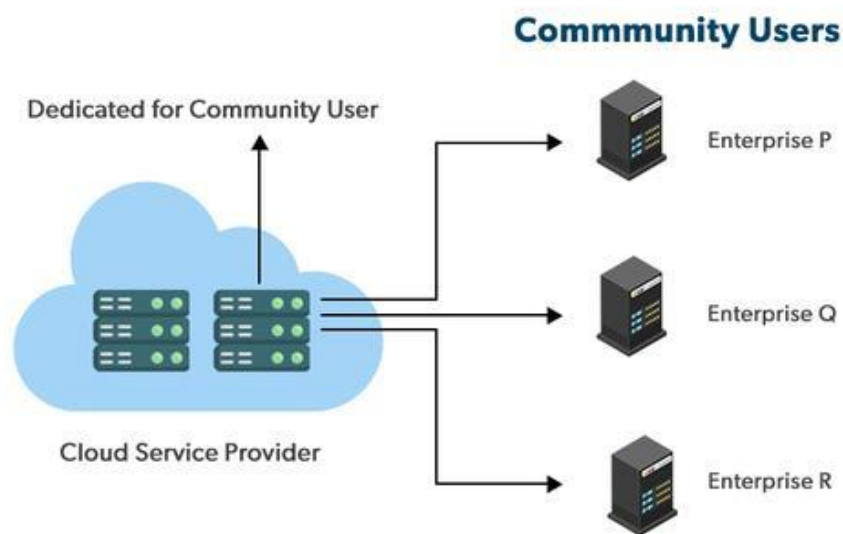


Public cloud



4. Community Cloud

- Community clouds are distributed systems created by integrating the services of different clouds to address the specific needs of an industry, a community, or a business sector. But sharing responsibilities among the organizations is difficult.
- In the community cloud, the infrastructure is shared between organizations that have shared concerns or tasks. An organization or a third party may manage the cloud.
- Examples: Cloud Sigma, Next cloud, Synology C2, OwnCloud, Stratos Cale



Introduction to Cloud Security

Cloud security refers to the protection of data, applications, and services stored and accessed through cloud computing platforms. It involves technologies, policies, and controls to safeguard cloud-based systems.

Importance of Cloud Security

- Protects sensitive data
- Prevents cyber attacks
- Ensures business continuity
- Maintains customer trust
- Helps meet legal and compliance requirements

Key Components of Cloud Security

1. Data Security

- Encryption (data is converted into secure format)
- Backup and recovery
- Data loss prevention

2. Identity and Access Management (IAM)

- Controls who can access data
- Multi-factor authentication (MFA)

3. Network Security

- Firewalls
- Intrusion detection systems (IDS)

4. Application Security

- Secure software development
- Regular updates and patching

Social site security

1. Introduction

Social site security is a part of cyber security that focuses on protecting users, their personal data, and online activities on social networking platforms. With the rapid growth of social media, millions of users share information daily, making these platforms a major target for cybercriminals.

2. Social Networking Sites Overview

Social networking sites are online platforms that allow users to connect, communicate, and share content globally.

Examples:

- Facebook
- Instagram

- Twitter (X)
- LinkedIn
- Snapchat

Features:

- Profile creation
 - Messaging and chatting
 - Sharing posts, photos, videos
 - Live streaming
 - Community groups
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3. Importance of Social Site Security

Social site security is important because:

- Protects personal and sensitive information
 - Prevents hacking and unauthorized access
 - Avoids identity theft and fraud
 - Ensures safe communication
 - Maintains user privacy and reputation
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4. Types of Threats in Social Media

4.1 Phishing Attacks

Attackers send fake messages or links to steal login credentials.

4.2 Malware Attacks

Malicious software spreads through links, ads, or downloads.

4.3 Identity Theft

Hackers use stolen personal data to impersonate users.

4.4 Social Engineering

Manipulating users into revealing confidential information.

4.5 Cyberbullying

Online harassment, threats, or abusive behavior.

4.6 Fake Profiles and Scams

Fraudsters create fake accounts to deceive users.

4.7 Data Breaches

Unauthorized access to large amounts of user data.

5. Common Security Risks

- Weak or reused passwords
 - Publicly visible profiles
 - Accepting unknown friend requests
 - Oversharing personal information
 - Using unsecured public Wi-Fi
 - Lack of awareness
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6. Security Measures and Best Practices

6.1 Strong Passwords

- Use a mix of letters, numbers, and symbols
- Avoid using personal information

6.2 Two-Factor Authentication (2FA)

Adds an extra layer of security by requiring a second verification step.

6.3 Privacy Settings

- Set profile visibility to private
- Control who can view posts and personal data

6.4 Be Careful with Links

- Avoid clicking unknown or suspicious links
- Verify the source before opening

6.5 Limit Personal Information Sharing

- Do not share phone number, address, or financial details publicly

6.6 Regular Updates

- Update apps and devices to fix security vulnerabilities

6.7 Log Out from Shared Devices

Always log out when using public or shared computers.

7. Role of Cyber Security in Social Media

Cyber security plays a major role in protecting social platforms by:

- Detecting cyber threats and attacks
 - Encrypting user data
 - Monitoring suspicious activities
 - Providing secure login systems
 - Preventing unauthorized access
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8. Laws and Regulations

Governments and organizations implement laws to protect users:

- IT Act (India)
- Data Protection Laws
- Cyber Crime Laws

These laws help punish cybercriminals and ensure user safety.

9. Advantages of Social Site Security

- Protects personal identity
 - Prevents financial loss
 - Ensures safe online interaction
 - Builds trust in digital platforms
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Protection against attack in social media

Update your software.

Up-to-date software systems are more resilient than outdated versions, which may be prone to having weaknesses. Updates can correct any flaws and weaknesses in the software, so having the latest version is optimal. Additionally, consider keeping software systems updated by investing in a patch management system.

Install a firewall.

Firewalls help prevent various attacks, such as backdoors and denial-of-service attacks. They work by controlling the network traffic moving through your system. A firewall will also stop any suspicious activity it deems potentially harmful to the computer.

Back up data.

When you back up data, you move it to a different, secure location for storage. This might involve using cloud storage or a physical device like a hard drive. In case of an attack, backing up your data allows you to recover any lost data.

Encrypt data.

Data encryption is a popular way to prevent cyberattacks. It ensures data is only accessible to those who have the decryption key. To successfully attack encrypted data, attackers often have to rely on the brute force method of trying different keys until they can guess the right one, making breaking the encryption challenging.

Use strong passwords.

You should have strong passwords to prevent attacks and avoid using the same passwords for different accounts and systems. Using the same password repeatedly increases the risk of giving attackers access to all your information. Regularly updating and using passwords that combine special characters, upper and lowercase letters, and numbers can help protect your accounts.

Securing wireless network

1. Enabling Two-Factor Authentication (2FA)

Two-factor [authentication](#) adds an extra layer of security to the login process. It requires users to enter both a username and password, as well as a code that is generated by an authenticator app. This makes it more difficult for someone to gain unauthorized access to the network.

To enable two-factor authentication, access the wireless router's configuration page and enable the feature. Be sure to download an authenticator app such as Google Authenticator or Authy and have it available when logging in.

2. Using A Strong Password

Using a strong password is one of the most important best practices for wireless network security. A strong password is at least eight characters long and includes a mix of upper- and lower-case letters, numbers, and symbols. Passwords should be changed regularly to ensure that they remain secure.

3. Using MAC Filtering

Using MAC filtering is another best practice for wireless network security. MAC addresses are unique identifiers assigned to devices that connect to a network.

By allowing only devices with specific MAC addresses to connect to the network, you can help to prevent unauthorized access. MAC filtering can be implemented by accessing the wireless router's configuration page and adding the MAC addresses of devices that are allowed to connect to the network.

4. Using A VPN

Using a VPN is another best practice for wireless network security. A VPN encrypts all traffic between a device and the VPN server, making it more difficult for someone to eavesdrop on the connection. This is especially important when using public Wi-Fi networks, as they are often less secure than private ones. Be sure to only [use VPNs from trusted providers](#) and make sure that employees are aware of the importance of using a VPN when working remotely.

5. Using A Firewall

Using a firewall is another best practice for wireless network security. A firewall helps to protect the network by blocking incoming traffic that is not authorized. This can be especially important in preventing attacks from malware and other malicious software.

To use a firewall, access the wireless router's configuration page and enable the feature. There are typically two types of firewalls: network-based and host-based.

Network-based firewalls are typically used in business environments, while host-based firewalls can be used on individual devices

Security Threats

Security threats are potential dangers that can harm computer systems, networks, or data. They can come from hackers, malware, or even human mistakes. Here's a clear and structured overview:

1. Malware (Malicious Software)

Software designed to damage or disrupt systems.

- **Virus** – Attaches to files and spreads when executed

In **cyber security**, a **virus** is a type of **malicious software (malware)** that is designed to **infect a computer system, replicate itself, and spread to other systems**, often causing harm to data, programs, or devices.

- **Worm** – Spreads automatically across networks

In **cyber security**, a **worm** is a type of **malicious software (malware)** that can **replicate itself and spread across computers and networks automatically**, without needing to attach to a file or program.

- **Trojan Horse** – Disguised as legitimate software

A **Trojan Horse** is a type of **malware** that **disguises itself as a legitimate or useful program** to trick users into installing it.

🔗 The name comes from the story of the **Trojan Horse**, where something harmful was hidden inside something that looked harmless.

- **Ransomware** – Locks data and demands payment

Ransomware is a type of **malware** that **blocks access to a computer system or encrypts files** and then demands **money (ransom)** to restore access.

- **Spyware** – Steals user information secretly

Spyware is a type of **malware** that **secretly monitors and collects information** about a user's activities without their knowledge.

2. Phishing Attacks

Fake emails, messages, or websites trick users into revealing sensitive data like passwords or bank details.

3. Password Attacks

Attempts to gain unauthorized access by cracking passwords:

- Brute force attack
- Dictionary attack
- Credential stuffing

A **password attack** is a method used by hackers to **steal, guess, or crack a user's password** to gain **unauthorized access** to systems, accounts, or data.

4. Denial of Service (DoS) Attack

Floods a system or website with traffic, making it unavailable to users.

- **DDoS (Distributed DoS)** uses multiple systems.

A **DoS (Denial of Service) attack** is a cyber attack where an attacker tries to **make a website, server, or network unavailable** by overwhelming it with too much traffic.

5. Man-in-the-Middle (MITM) Attack

Attacker secretly intercepts communication between two parties (e.g., public Wi-Fi hacking).

6. Insider Threats

Employees or authorized users misuse access intentionally or accidentally.

7. SQL Injection

Attackers insert malicious SQL code into a database query to access or modify data.

SQL Injection (SQLi) is a type of cyber attack where an attacker inserts **malicious SQL code** into a website's input fields (like login forms) to **access, modify, or delete database data**

8. Zero-Day Exploits

Attacks that exploit unknown vulnerabilities before developers can fix them.

9. Social Engineering

Manipulating people to reveal confidential information (e.g., pretending to be a trusted person).

Social Engineering is a cyber attack technique where attackers **manipulate people (not systems) to reveal confidential information** like passwords, OTPs, or bank details.
